



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2024

Paper: Categorical Data Analysis

Course Code: STAT-416

Roll No.

Time: 3 Hrs



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (5x6=30)

- Differentiate among the different types of categorical variables with the help of examples.
- Discuss the applications and limitations of chi-square distribution.
- What is the purpose of Kendall's Tau-b test? Explain its testing procedure.
- Define the Fisher's Exact Test. When is it appropriate to use this test instead of the chi-square test?
- A random sample of 30 adults was classified according to gender and the number of hours they watch television during a week. Use phi-coefficient to measure the degree of association between the two attributes.

Classes	Male	Female
More than 25 hours	9	5
Less than or equals to 25 hours	7	9

Answer the following questions. (3x10=30)

- Q.2** A bakery observes the number of customers arriving at their store every hour for a week. Fit an appropriate distribution to the given data and test the hypothesis of goodness-of-fit.

No. of Customers	0	1	2	3	4	5	6	More than 6
Frequency	13	30	45	42	25	10	4	0

- Q.3** The following contingency table displays the count of people experiencing joint pain before and after a certain treatment, and those who don't experience the pain before and after the treatment. Apply McNemar's test explore whether the treatment worked or not. Interpret your findings.

Experienced Pain Before Treatment	Experienced Pain After Treatment	
	Yes	No
	Yes 101	No 121
No	160	33

- Q.4** A public health study investigates whether consuming sugary drinks is associated with an increased risk of obesity. Researchers collected data from 500 participants and classified them into two groups based on their sugary drink consumption. Group A (Consumes sugary drinks regularly): 150 participants, 90 of whom are obese. Group B (Does not consume sugary drinks regularly): 350 participants, 70 of whom are obese.

- Calculate the Relative Risk (RR) of obesity for individuals who consume sugary drinks regularly compared to those who do not.
- Calculate the Odds Ratio (OR) for obesity between the two groups.
- Discuss the interpretation of both RR and OR in the context of the association between sugary drink consumption and obesity.